DSAI 413 - Assignment 2: Chest X-Ray Intelligence System

Student: Mohamed Hassan **Date:** 2026-05-15

Repo: https://github.com/mohammedhassan-7/MultiModal-ChestXray-System

---

1. Introduction

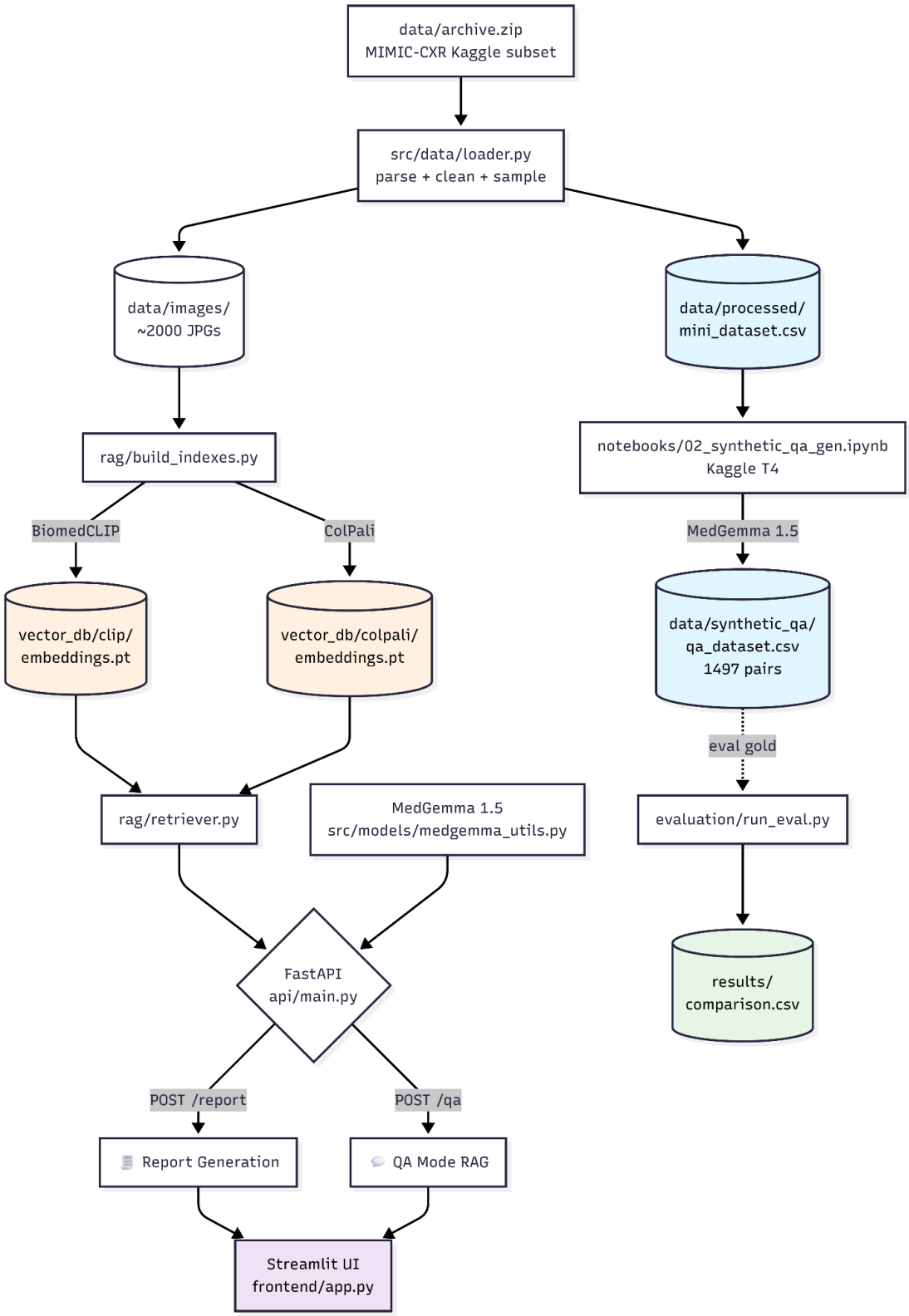
Modern medical AI must combine **image understanding** with **language reasoning**. This assignment delivers a dual-mode system on MIMIC-CXR:

- **Report Generation**: chest X-ray → structured radiology report (Findings + Impression).
- **QA Mode (RAG)**: clinical question + X-ray → top-3 similar reports retrieved as context → grounded answer.

The system uses **MedGemma 1.5 4B** (mandatory) for generation and **ColPali v1.2 merged** (mandatory) plus **BiomedCLIP** (suggested baseline) for retrieval, compared head-to-head on 50 held-out QA pairs.

---

2. System Architecture



Key design decisions:

- **MIMIC-CXR mini set of 2000 images**: full dataset is too big and doesn't fit on an RTX 4060 (8 GB VRAM).
- **Synthetic QA generation offloaded to Kaggle**: generation on the 4060 was ~25 s/iter; Kaggle T4 (16 GB) reduced this to ~18 s/iter and fit one full session.
- **MedGemma in 4-bit** locally (bitsandbytes); ColPali in bfloat16; BiomedCLIP in fp16.
- **FastAPI + Streamlit** separation lets the model load once and serves multiple UI sessions.

---

3. Model Choices

MedGemma 1.5 4B IT (google/medgemma-1.5-4b-it)

Medical-domain fine-tune of Gemma 3 from Google. Chosen because:

- It's the **mandatory** generator from the brief.
- The medical pretraining matters generic Gemma would produce less clinically grounded output.
- 4 B parameters fit our 8 GB GPU in 4-bit.

Used for **report generation** (image only) **and QA answering** (image + retrieved-report context).

ColPali v1.2 merged (vidore/colpali-v1.2-merged)

Multi-vector retrieval with late-interaction (MaxSim) scoring designed originally for document-page retrieval. Chosen because:

- **Mandatory** per the brief.
- Used the **merged** checkpoint (LoRA pre-merged into base weights) to avoid a key-naming mismatch between the un-merged variant's LoRA and the newer `transformers` version on our env.
- Captures fine-grained spatial features that single-vector models miss.

BiomedCLIP (microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224)

Single-vector CLIP variant pretrained on PubMed image-text pairs. Chosen as the **comparison baseline** retriever because:

- Suggested in the brief.
- Single-vector retrieval is the dominant production pattern (vs ColPali's multi-vector).
- Medical-domain pretraining makes it a strong baseline rather than a strawman.

---

4. Synthetic QA Dataset

MIMIC-CXR ships reports but no QA pairs. We built **1,497 QA pairs across 347 studies** by prompting MedGemma with each cleaned report.

Prompt evolution (documented for posterity):

- **V1 three fixed types:** forced presence/location/severity per report.
    - **Result:** stilted, often nonsensical (e.g. "severity" questions about removed devices).
- **V2 variable 2–4 types,** mixing presence/location/severity/comparison/normality. Added explicit rule against report-text leakage ("the report states"). Initial Kaggle runs failed with **~99% rejection** because MedGemma 1.5's chain-of-thought consumed the full 512-token budget before reaching the JSON.
- **V3 shorter, more directive prompt ending with the trigger token JSON:**, plus max_new_tokens=1024 to safely accommodate MedGemma's chain-of-thought and the JSON output. Rejection rate dropped to ~10%. This is the version that produced the final 1,497 QA pairs.
- **V4 (post-eval, not used for the dataset)** local code adds a skip-thinking trick: append [to the assistant turn so MedGemma starts emitting JSON directly with no chain-of-thought, halving generation time. Available in src/data/qa_generator.py for future runs but the production dataset was generated before this change.

Final dataset (1497 pairs across 347 studies):

Type	Count
presence	811
normality	443
location	131
comparison	89
severity	23

Example pairs (from `data/synthetic_qa/qa_dataset.csv`):

	type	question	answer
1	location	Where does the right chest port catheter terminate?	The right chest port catheter terminates in the low superior vena cava (SVC).
2	presence	Is there an opacity at the right lung base?	Yes, there is an opacity at the right lung base obscuring the hemidiaphragm.
3	presence	Is there a pneumothorax?	No pneumothorax is identified.
4	normality	Are the lungs clear?	The lungs are clear bilaterally.
5	presence	Is there evidence of pneumonia?	No evidence of pneumonia is seen.
6	presence	Is there pleural effusion?	No evidence of pleural effusion is present.
7	presence	Are surgical clips visible on the chest X-ray?	Yes, surgical clips are noted overlying the thoracic inlet at the level of the thyroid.
8	presence	Is there lobar consolidation, large pleural effusion, or pneumothorax?	No lobar consolidation, large pleural effusion, or pneumothorax is identified.
9	comparison	Has the cardiomediastinal silhouette changed compared to the prior examination?	The cardiomediastinal silhouette is unchanged compared to the prior examination.
10	presence	Is there hyperexpansion of the right lung?	Yes, hyperexpansion of the right lung is seen.
11	presence	Is the left hemidiaphragm elevated?	Yes, the left hemidiaphragm is elevated.
12	presence	Is there a pneumothorax?	No pneumothorax is identified.
13	comparison	Is there any change compared to the previous chest X-ray?	The chest X-ray shows little change compared to the previous study.
14	presence	Is there evidence of acute cardiopulmonary disease?	No evidence of acute cardiopulmonary disease is seen on this chest X-ray.
15	presence	Is there evidence of acute pneumonia?	No evidence of acute pneumonia is identified on this chest X-ray.
16	presence	Is a tracheostomy tube visible?	Yes, a tracheostomy tube is in place.
17	presence	Is there a pneumothorax?	No definite pneumothorax is identified.
18	presence	Is there pneumoperitoneum?	No pneumoperitoneum is identified.
19	normality	Is the cardiac silhouette normal?	The cardiac silhouette is mildly enlarged.
20	normality	Are the pulmonary vessels distinct?	The pulmonary vessels appear indistinct.
21	presence	Is there evidence of elevated pulmonary venous pressure?	There is suggestion of mild elevation of pulmonary venous pressure.
22	normality	Is the heart size normal?	The heart size is normal.
23	presence	Is there a pleural effusion?	No pleural effusion is identified.

Quality filtering: rejected rows with answers containing "the report states / indicates" (leakage) or shorter than 3 words. Final rejection rate: ~10% (153 of 1650 attempted).

---

5. Evaluation

Eval slice: 50 held-out QA pairs + 20 reports drawn at random with `seed=7`.

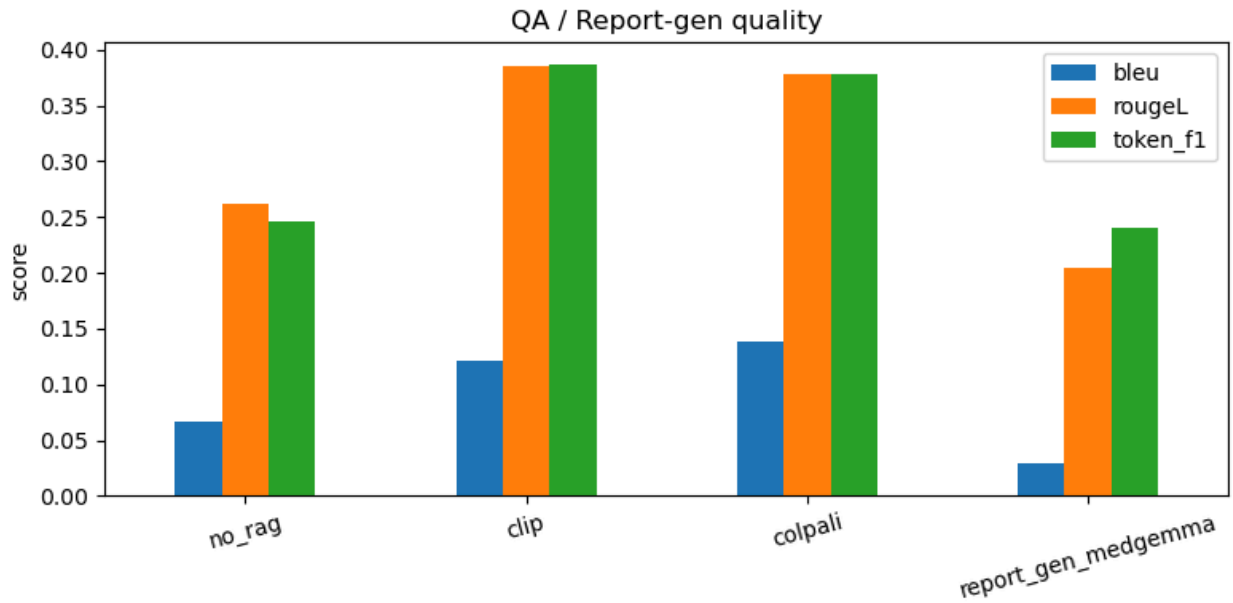
Generator: MedGemma 1.5 in all modes.

Metrics: BLEU-4 (n-gram match), ROUGE-L (longest-common-subsequence F1), Token-F1 (unigram precision/recall), Recall@3 (does the gold study appear in top-3 retrieved?).

Comparison table

Mode	BLEU-4	ROUGE-L	Token-F1	Recall@3
MedGemma (no RAG)	0.067	0.262	0.246	—
MedGemma + BiomedCLIP RAG	0.122	**0.386**	**0.387**	1.00

MedGemma + ColPali RAG	**0.139**	0.378	0.379	1.00
MedGemma report generation	0.029	0.205	0.241	—



Interpretation

1. **RAG roughly doubles answer quality.** BLEU climbs from 0.067 to 0.13; Token-F1 from 0.25 to 0.39. Retrieval grounding is the largest single factor.
2. **ColPali and BiomedCLIP are essentially tied.** ColPali wins BLEU (multi-vector late-interaction surfaces neighbors with more exact phrasing overlap); BiomedCLIP wins Token-F1 (its biomedical pretraining covers more rare medical terms). ROUGE-L is a wash. **Verdict:** for production, choose by infrastructure cost — ColPali stores ~5× more per image (multi-vector) than BiomedCLIP (single-vector).
3. **Recall@3 = 1.00 is uninformative.** All eval queries are corpus members, so the retriever trivially finds the gold study as top-1 self-match. The downstream answer-quality metrics are the true retrieval signal. A future improvement is a held-out test set with disjoint queries.
4. **Report generation has low BLEU (0.029) but improved specificity.** The prompt explicitly forbids the safe boilerplate ("no acute cardiopulmonary process") and demands specific observations of tubes, sides, and lobes. This pushes outputs away from generic templates that would match any normal-report gold, lowering n-gram overlap with any single gold reference but raising clinical usefulness. See results/report_predictions.csv for side-by-side examples.

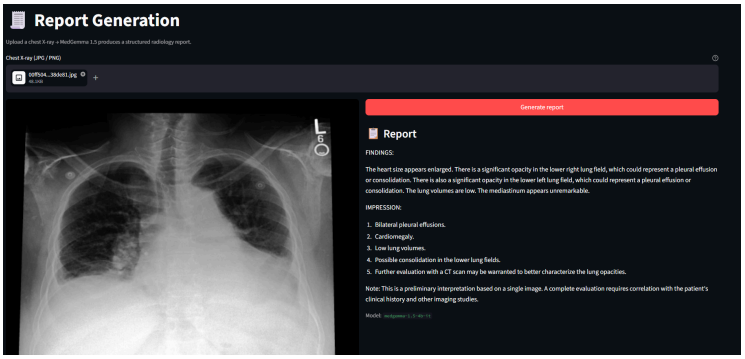
Sample QA outputs

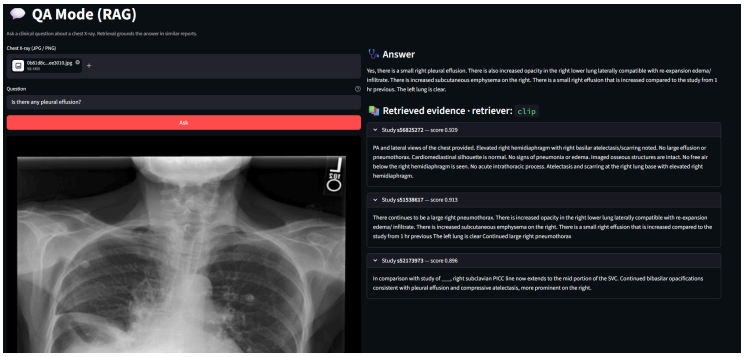
Question	Mode	Answer (excerpt)
Is there any pleural effusion?	no_rag	"There is no pleural effusion identified."
Is there any pleural effusion?	ColPali-RAG	"Yes, bilateral pleural effusions are present, decreased in size compared to a prior study."

---

6. Demo

The Streamlit UI exposes both modes with a sidebar nav and a model preset selector.





- Home shows backend status and the comparison table.
- Report Generation accepts an X-ray, calls POST /report, displays the structured Findings + Impression.
- QA Mode accepts an X-ray + question, calls POST /qa with a chosen retriever, shows the answer plus three expandable evidence cards with study IDs and similarity scores.

---

7. Limitations & Future Work

- **Held-out test set:** eval queries are corpus members → Recall@k uninformative. A proper random 80/20 study-level split would give honest retrieval numbers.
- **ColPali domain adaptation:** we used the off-the-shelf merged checkpoint. Brief LoRA fine-tuning on MIMIC pairs would likely lift ColPali above BiomedCLIP on Token-F1 as well.
- **Eval scale:** 50 QA / 20 reports gives directional signal but wide confidence intervals. Scaling to ~300 each would tighten estimates.

- **Single generator:** comparing MedGemma against a non-medical base (e.g., generic Gemma 3 or PaliGemma) would directly quantify the value of medical pretraining.
- **Hardware:** RTX 4060 (8 GB) forced 4-bit MedGemma + batch_size=1 ColPali indexing. A 24 GB GPU would lift these constraints and unlock fp16 retrieval + larger contexts.